Negar Neda

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Education

New York University, Tandon School of Engineering *Sep. 2021 -2026*
PhD in Electrical and Computer Engineering, Advisor: Brandon Reagen
Thesis title: Accelerating Homomorphic Encryption Implementation

University of Tehran (UT), Tehran, Iran *Sep. 2018 - Feb. 2021*
M.Sc. in Computer Architecture, GPA: 3.63/4
Thesis title: FPGA-based Multi-precision Accelerator for Deep Neural Networks

Amirkabir University of Technology (AUT), Tehran, Iran *Sep. 2014 - Sep. 2018*
B.Sc. in Computer Architecture, GPA: 3.62/4
Thesis title: Implementation of a Tracking System Using LoRaWAN Protocol

Current Research

I design **high-performance hardware accelerators** for privacy-preserving and compute-intensive workloads, focusing on **FHE** and **LLMs**. My current work proposes a **chiplet-based architecture** for accelerating TFHE bootstrapping with low latency and high throughput. I also work on optimizing matrix multiplication for LLMs on encrypted data. My research interests include **domain-specific architectures**, **secure hardware design**, and **HPC** for **cryptography** and **machine learning**.

Publications

· "Network and Compiler Optimizations for Efficient Linear Algebra Kernels in Private Transformer Inference (Invited Paper)", International Conference on Computer-Aided Design (ICCAD), **N. Neda**, K. Garimella, A. Ebel, N. Kumar Jha, B. Reagen, 2025

· "CiFlow: Dataflow Analysis and Optimization of Key Switching for Homomorphic Encryption", IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS), **N. Neda**, A. Ebel, B. Reynwar, B. Reagen, 2024.

· "Quantifying the Overheads of Modular Multiplication", IEEE International Symposium on Low Power Electronics and Design (ISLPED), D. Soni, M. Nabeel, **N. Neda**, et al., 2023.

· "Towards fast and scalable private inference", Proceedings of the 20th ACM International Conference on Computing Frontiers, J. Mo, K. Garimella, **N. Neda**, A. Ebel, B. Reagen, 2023.

· "RPU: The Ring Processing Unit", IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS), **N. Neda**, D. Soni, et al., 2023.

· "Multi-Precision Deep Neural Network Acceleration on FPGAs", Asia and South Pacific Design Automation Conference (ASP-DAC), **N. Neda**, et al., 2022.

Research Experience

Graduate Research Assistant, *New York University* 2021 - Present
· Doing research on the design of **hardware accelerators** for **privacy-preserving machine learning** algorithms, with a focus on **Homomorphic Encryption** (HE), including accelerating **LLM inference** on encrypted data.

Research Assistant, *University of Tehran* 2018 - 2021
·Implemented an FPGA-based Multi-precision accelerator for DDNs.

Undergrad Researcher, *Amirkabir University of Technology* 2017 - 2018
· Designed Amirkabir University of Technology IoT Gateway.

Teaching Experience

· Computing Systems Architecture · Computer Aided Digital System Design · Logic Circuit Laboratory
· Digit Design Automation

Selected Projects

Accelerating MatMult in LLMs for Encrypted Data in FHE *2025 – ongoing*
· Exploring **Hardware-Algorithm co-design** & **GPU** kernels to accelerate **LLM inference** on HE-encrypted data (CKKS), focusing on high-throughput MatMul and HE-friendly nonlinear layers.

Chiplet-Based Accelerator for TFHE *2024 – ongoing*
· Designed a **chiplet-based** accelerator for **TFHE** programmable bootstrapping, achieving up to **2×** speedup and throughput without HBM, with a compact $\mathbf{13\,mm^2}$ chiplet area.

Optimized Key-Switching Dataflow for Efficient Homomorphic Encryption *2023*
· Improved data reuse by strategically scheduling key-switching operations in HE, achieving **4×** speedup and **12×** SRAM savings by minimizing off-chip data movement.

Vector-based ISA & Microarchitecture for NTT Acceleration *2022*
· Developed a **vector ISA** and custom hardware for accelerating RLWE-based workloads, achieving **1485×** speedup for **NTT** over CPU using modular arithmetic support and parallel processing.

5-Stage MIPS Processor with Pipelining *2021*
· Built a pipelined **MIPS processor** in C++ with a 2-bit branch predictor and a direct/fully associative cache implementation.

Multi-Precision Accelerator for DNN on FPGA *2021*
· Designed a dynamic bit-width multiplier for **DNNs** on **FPGA**, enabling 3–15× throughput by dynamically adjusting bit-width to match precision needs.

HONOR & AWARDS

· Awarded a Student Travel Grant to present our work at ISPASS 2023 & 2024
· Earned **Future Leader Fellowship** at NYU Tandon. 2022-2024
· **Ranked Top 3 in terms of GPA**, among Computer Architecture Students in AUT 2019

Relevant Courses

· Computing Systems Architecture · Intro to Cryptography · Hardware Security · Machine Learning
· Neural Networks · Multicore Embedded Systems · VLSI Systems Design

Technical skills

Programming	Python(Tensorflow, PyTorch), Rust, C/C++, VHDL, System/Verilog, CUDA, OpenMP
Tools	Visual Studio, Synopsys Design Compiler, Vivado Design Suite, PSPICE, HSPICE, Modelsim, MATLAB, Jupyter Notebook, Git

Talks

· "CiFlow", Columbia Computer Architecture Day 2025
· "CiFlow: Dataflow Analysis & Optimization of Key Switching for HE", ISPASS 2024
· "RPU: The Ring Processing Unit", ISPASS 2023
· "Summary of RPU: The Ring Processing Unit ", Princeton Computer Architecture Day 2023
· "A Novel Vector ISA for Accelerating Homomorphic Encryption", TECHCON 2022

Outreaches

· Session Chair, NYU Computer Architecture Day 2024
· Staff Member, International Symposium on Computer Architecture (ISCA), NY 2022